Similarly to `bubbleUp`, also `pushDown` can be improved by simply keeping current element in a temporary variable, and avoiding swaps at each iteration.

Listing 2.5 shows the final improved version of the method, we encourage the reader to work our example out line by line, similarly to what we have done in the previous section for `bubbleUp`, to have a better grasp of how it works.

Now that we have defined all the helper methods we need, we can finally implement the API methods: as a spoiler, you will notice how all the logic about priority and max-heap vs min-heap is encapsulated into `bubbleUp` and `pushDown`, so by defining these two helper methods we greatly simplify our life whenever it comes to adapting our code to different situations.

2.6.3 Insert

Let's start with insertion: listing 2.6 describes the pseudocode for inserting a new `(element, priority)` pair into a heap.

As mentioned at the end of the previous section, the insert method can be written by leveraging our helper methods, and as such its code is valid independently on whether we use a min-heap or a max-heap and on the definition of priority.

Listing 2.6 The insert method

```
function insert(element, priority)
```